UNIVERSAL PROJECT HANDBOOK | COMPLETE EXPLAINER GUIDE | DUAL ENGINE: SQLITE + SUPABASE | 99/99 PYTEST PASSED

QueryMind 2.0: The Complete Project Explainer & Architecture Handbook

Purpose: A complete, self-explanatory guide for any person to fully understand QueryMind 2.0

Target System: QueryMind 2.0 (Branch: `querymind-2.0`) | **Date:** October 2026

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1. The Big Picture: What is QueryMind 2.0?

> THE SIMPLE METAPHOR ANYONE CAN UNDERSTAND
>
> **Think of QueryMind 2.0 like Siri or Google Assistant, but exclusively built for enterprise relational SQL databases.**
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> Normally, querying a college database requires writing complex SQL code full of joins, group-by clauses, and foreign keys. Ordinary people—like teachers, students, or college clerks—cannot do this.
>
> With QueryMind 2.0, you simply click the microphone on your screen and **speak in plain English** (for example: *"Find students with attendance below 75%"* or *"Show average marks by subject"*).
>
> The system captures your voice, converts it into clean SQL, checks that the query is safe, shows it to you on screen so you can review it, runs it against the real database, and displays the matching records in an interactive table—all in a fraction of a second!

Why not just use ChatGPT or a regular AI?

If you simply connect an unconstrained AI (like ChatGPT) to a database, it is **extremely dangerous** for three reasons:

1. **Hallucinations:** Generative AI frequently makes up table or column names that don't exist in the database, causing crashes.
2. **Destructive Mistakes:** An AI can easily output `DROP TABLE students;` or `DELETE FROM marks;` when confused, accidentally wiping out college records.
3. **SQL Injection Attacks:** Hackers can trick generative AI by speaking malicious instructions to bypass passwords and steal sensitive data.

QueryMind 2.0 solves this completely: It does NOT use unconstrained generative AI. Instead, it uses a **deterministic, schema-aware pipeline** governed by mathematical syntax trees. It is physically impossible for QueryMind 2.0 to delete data, hallucinate fake tables, or run unauthorized queries.

2. How the System Works: Step-by-Step Flow

Whenever someone asks: *"What actually happens under the hood when I speak into QueryMind 2.0?"*, here is the exact 8-step journey:

STEP 1 **Audio Capture via Browser Microphone**

The user clicks the mic button in the web browser. The browser-native W3C Web Speech API captures acoustic sound waves and converts spoken audio into text in real time, displaying an active pulsing audio waveform animation.

STEP 2 **Colloquial Text Normalization**

Spoken English often doesn't match SQL. Our normalizer automatically standardizes spoken terms:
- Spoken words like *"seventy five"* → converted to numeric `75`.
- *"Percent"* or *"percentage"* → converted to `%`.
- *"Division a"* → standardized to uppercase literal `'A'`.

STEP 3 **The Intent Classifier (The "Bouncer" at the Door)**

Before any SQL is generated, the **Intent Classifier** inspects the utterance. If someone says something non-database like *"hello how are you"* or *"tell me a joke"*, the bouncer **halts execution immediately**. Zero SQL is generated, no database is touched, and helpful suggestions appear on screen.

STEP 4 **Schema-Aware SQL Generation**

If the intent is a valid database query, our template engine checks the connected database schema. It knows that student names are in `students` and percentages are in `attendance`, so it automatically writes the correct `INNER JOIN` and `WHERE` clauses.

STEP 5 **Mathematical AST Security Validation (SQLGlot)**

The generated query is fed into **SQLGlot**, an in-memory Abstract Syntax Tree parser:
- It verifies the query is 100% read-only (only `SELECT` is allowed; `DROP`, `DELETE`, `UPDATE` are blocked).
- It rejects multi-statement semicolon injections (`SELECT 1; DROP TABLE`).
- It automatically injects a defensive `LIMIT 100` row ceiling so large queries never crash the server.

STEP 6 **Pre-Execution Review in the Workspace**

A fundamental safety rule: Stopping speech NEVER automatically runs a query! The synthesized SQL is placed into an interactive code editor with distinct badges (Speech Recognized, Intent Validated, AST Validated). The human user must review the query and explicitly click **[Run Query]**.

STEP 7 **Database Execution (PostgreSQL or SQLite)**

Our unified `DatabaseAdapter` sends the query to the active engine—either cloud **Supabase PostgreSQL 15+** (with kernel-level Row-Level Security) or local **SQLite** (in WAL mode). The database executes the query in just **2 to 4 milliseconds**.

STEP 8 **Results Display & Execution Plan Profiling**

The query results are rendered in an interactive table with sorting, row counts, and latency timers. Users can click the blue **[Explain Plan]** drawer to see the database execution plan, showing whether the database did a fast B-Tree index search or a full table scan.

3. The Database Design (What data is stored?)

QueryMind 2.0 is built on an academic institutional database with strict relational modeling:

Table Name	Primary Key	Foreign Key References	Purpose & Relational Attributes
`students`	`student_id` (INT)	None	Stores student profile: `name`, `division` (e.g. 'A', 'B'), and `email`.
`subjects`	`subject_id` (INT)	None	Stores academic subjects: `subject_name` (Math, Physics, CS) and `credits` (weighting).
`marks`	`mark_id` (INT)	`student_id` → `students` `subject_id` → `subjects`	Transactional scores table: stores numerical `marks` (0 to 100). Cascades on deletion. Composite unique constraint prevents duplicate scores.
`attendance`	`attendance_id` (INT)	`student_id` → `students` `subject_id` → `subjects`	Tracks student presence: stores `attendance_percentage` (0 to 100%). Used for defaulter radar queries. Cascades on deletion.
`app_users`	`user_id` (INT)	None	User accounts with roles: `student`, `faculty`, and `admin`. Passwords hashed with SHA-256. Masked from student queries.
`query_logs`	`id` (INT)	`user_id` → `app_users`	Audit log recording question text, synthesized SQL, execution status, latency (ms), and input mode (`voice` vs `text`).

Why is this in Third Normal Form (3NF)? (Simple Explanation)

Normalizing a database means designing tables so that information is never duplicated unnecessarily. If you duplicate information, updating a student's name in one place might leave it outdated in another (an update anomaly).

- **First Normal Form (1NF):** All values are single atomic pieces of data (no lists, no comma-separated text). Every table has a unique Primary Key.
- **Second Normal Form (2NF):** In tables with composite keys like `marks` (where each row represents a student and a subject), every column depends on the *whole* key. Course metadata (like subject credits) is separated into the `subjects` table so it isn't repeated for every student.
- **Third Normal Form (3NF):** No transitive dependencies ($X \to Y \to Z$). A student's division depends on their `student_id`. Therefore, division is stored in `students`, NOT inside `marks`. `marks` only stores the score and foreign keys!

4. The Story of the Critical Bug: The Greeting Bug Autopsy

THE REAL ENGINEERING CHALLENGE WE HAD TO SOLVE

What was the bug?

When we tested the microphone with ordinary conversational greetings like:

"Hello how are you"

The system unexpectedly produced a real database query:

```
SELECT * FROM "exam_scores" LIMIT 50
```

...and it displayed bright green badges saying: *"SQL generated successfully"* and *"AST Validated • Read-Only SELECT • Safe"*!

Why was this terrible?

The user never asked for database records. Synthesizing arbitrary queries for random talk proves that the system wasn't truly understanding intent.

The Investigation & Root Cause:

We traced the code across the entire backend and identified three distinct flaws:

1. **The Hidden Fallback in Code:** Inside `nlp/sql_generator.py` at line 402, the function `generate_sql_for_dataset()` contained an unconditional fallback clause: whenever none of the pattern keywords matched, it returned `SELECT * FROM "{table_name}" LIMIT 50` with `matched: True`!
2. **Missing Intent Gating:** The Flask API gateway sent every transcript straight into SQL generation without first asking: *"Is this person actually asking about the database?"*
3. **Frontend Stale State Persistence:** In `static/js/voice_sql.js`, when a query failed, the code editor was not cleared, leaving previous queries executable.

How We Fixed It Permanently:

1. **Removed the Arbitrary Fallback:** In `nlp/sql_generator.py`, unmatched queries now strictly return `matched: False, sql: None`.
2. **Created `nlp/intent_classifier.py`:** We introduced a 5-state Intent Classifier that categorizes queries before SQL generation:
   - `OUT_OF_SCOPE`: Greetings (*"hello"*, *"tell me a joke"*) → Halted. Editor cleared. Suggests valid DB questions.
   - `AMBIGUOUS_QUERY`: Vague requests (*"show data"*) → Halted. Prompts user to pick a table.
   - `UNSUPPORTED_OPERATION`: Destructive commands (*"delete all students"*) → Halted. Explains read-only policy.
   - `EMPTY_INPUT`: Silence / noise → Prompts user to speak a question.
   - `DATABASE_QUERY`: Legitimate query → Allowed to generate SQL.
3. **Frontend State Reset:** `static/js/voice_sql.js` now wipes the SQL editor, hides AST badges, and presents helpful suggestion chips whenever input is invalid.
4. **40 Regression Tests:** We created `tests/test_intent_regression.py` with 40 automated tests proving that greetings can never generate SQL.

5. How to Explain QueryMind 2.0 to Any Audience

IF EXPLAINING TO A COLLEGE EXAMINER OR PROFESSOR

"QueryMind 2.0 is an academic and enterprise voice-to-RDBMS query platform. We enforce strict relational normalization (3NF), deterministic 5-state intent classification, and mathematical AST query governance using SQLGlot to guarantee read-only execution. It features a dual-engine architecture supporting cloud Supabase PostgreSQL with Row-Level Security alongside SQLite in WAL mode, backed by an automated 99-test pytest suite."

IF EXPLAINING TO A FRIEND, JUNIOR, OR NON-TECHNICAL PERSON

"Imagine you're a college teacher and you want to know which students have low attendance. Instead of asking a programmer to write a SQL database query for you, you just open our website, press the microphone, and say: *'Find students with attendance below 75%'*.

Our system listens to your voice, turns it into the exact SQL code, checks that it's safe, and displays the students on your screen in two seconds. And if you say something silly like *'hello how are you'*, it knows you're not asking about the database and tells you politely what to ask instead!"

6. Top 10 Questions Anyone Will Ask (With Simple Answers)

Question	Simple, Clear Answer
1. Can a user delete the database by speaking "Delete all students"?	**No, impossible.** First, our Intent Classifier flags it as `UNSUPPORTED_OPERATION`. Second, our SQLGlot AST validator physically rejects any query that isn't a read-only `SELECT`.
2. Does it run the query immediately when I stop talking?	**No, never.** Audio stop only transcribes your speech. The generated SQL sits in an editor for you to review and inspect. You must explicitly click **[Run Query]** to execute it.
3. Why do you have two databases (SQLite and PostgreSQL)?	SQLite runs locally on your laptop with zero setup (great for testing). Supabase PostgreSQL runs in the cloud (great for real colleges with thousands of users and privacy security).
4. How do you prevent student users from seeing other students' marks?	Using Supabase **Row-Level Security (RLS)**. The database kernel checks who is logged in and automatically hides rows that don't belong to that student.
5. What is the [Explain Plan] drawer on the screen?	It shows how the database found your data. It tells you whether the database had to read every single row (Full Table Scan) or if it jumped straight to the answer using a fast B-Tree index.
6. How many automated tests does the project have?	We have **99 automated pytest tests** that run in 3.99 seconds with a 100% pass rate.
7. What happens if there is background noise or silence?	The Intent Classifier tags it as `EMPTY_INPUT` and asks you to speak a question about connected database entities.
8. What is the ER diagram on the website?	It's an interactive diagram of all tables and relationships generated dynamically using **Mermaid.js** by reading the database's live foreign keys.
9. How does it understand numbers like "seventy five percent"?	We built a spoken-number normalizer that translates colloquial words into numeric digits (`75%`) before generating SQL.
10. Can I upload my own custom data?	Yes! Administrators can upload any CSV file in the Admin Dashboard, which automatically creates a new database table and lets you query it by voice.

7. Live Demonstration Cheat-Sheet: 5 Queries to Show Anyone

If you ever have to demonstrate QueryMind 2.0 live on a laptop, speak these 5 queries in this exact order to show off all features:

1. **Query 1 (Show Intent Gating & Greeting Fix):**
   Click the mic and say: *"Hello how are you"*.
   Show them: The SQL editor stays completely blank! The system shows status `out_of_scope` and presents clickable suggestions. Explain that QueryMind 2.0 never generates random queries for chit-chat.
2. **Query 2 (Show Ambiguous Clarification):**
   Click the mic and say: *"Show me the data"*.
   Show them: The system identifies this as `AMBIGUOUS_QUERY` and asks: *"Which table would you like to query? (students, marks, attendance)"*.
3. **Query 3 (Show Real Database Voice Query & Normalization):**
   Click the mic and say: *"Students with attendance below 75 percent"*.
   Show them: Show that spoken *"seventy five percent"* became `75%`, and show the synthesized SQL with `INNER JOIN attendance`. Show the green **AST Validated** badge.
4. **Query 4 (Show Execution & Performance Profiler):**
   Click **[Run Query]**, then click the blue **[Explain Plan]** button.
   Show them: The results table with student names and attendance percentages, and the execution tree showing B-Tree index seeks vs table scans.
5. **Query 5 (Show Live Dynamic Mermaid ER Diagram):**
   Click on **Database Explorer** in the top navigation bar.
   Show them: The live Mermaid.js ER diagram rendered dynamically from SQLite catalog pragmas with 1:N cardinality connectors.

> **SUMMARY CHECKLIST**
> - **Codebase:** Verified on GitHub branch `querymind-2.0`
> - **Tests:** 99/99 passed in 3.99 seconds
> - **Architecture:** Deterministic Intent Classifier + SQLGlot AST + Dual Engine (SQLite + Supabase)
> - **Key Principle:** Safety first—microphone never auto-executes, and queries are always human-reviewed.